



FPT UNIVERSITY

Faculty of Artificial Intelligence

Capstone Project Report

An Offline Intelligent Vietnamese Input Method Based on Deep Learning

NextKey		
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Capstone Project code	AIP491	

Da Nang, August 2026

Abstract

This capstone project, NextKey, addresses the cognitive and physical challenges of typing Vietnamese on mobile devices caused by traditional rule-based methods like Telex and VNI. We propose an intelligent, end-to-end sequence-to-sequence Deep Learning system that allows users to type continuous, unaccented, and unsegmented Latin characters (similar to Pinyin). The model jointly performs word segmentation and diacritic restoration to predict the fully formatted Vietnamese sentence in real time. To ensure seamless interaction and absolute data privacy, the architecture is heavily optimized using Knowledge Distillation and Quantization for offline, low-latency execution on edge devices. This novel input method aims to significantly improve typing ergonomics, maximize keystroke savings, and provide an accessible language scaffold for all users.

Keywords: Vietnamese Input Method, Diacritic Restoration, Word Segmentation, Sequence-to-Sequence Modeling, Edge Computing, Offline Inference.

Acknowledgments

We would like to express our deepest gratitude to our academic supervisor, Ass.Prof Nguyen Luong Vuong, for his invaluable guidance, continuous support, and insightful feedback throughout the duration of this capstone project. His expertise has been instrumental in shaping our research methodology and system design.

We also extend our sincere thanks to the Faculty of Artificial Intelligence at FPT University Da Nang Campus for providing an excellent academic environment and the necessary resources to conduct this research.

Finally, we are deeply grateful to our families, friends, and peers for their unwavering encouragement and moral support during the intensive development phases of this project.

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Definition and Acronyms

AI Artificial Intelligence

ML Machine Learning

I. Project Introduction

1. Overview

1.1. Project Information

- **Project Title:** An Offline Intelligent Vietnamese Input Method Based on Deep Learning
- **Project Code:** AIP491 (Capstone Project in Artificial Intelligence)
- **Academic Institution:** Faculty of Artificial Intelligence, FPT University Da Nang Campus
- **Team Members:**
 - Leader: Tran Dinh Gia (Student ID: SE181550)
 - Member 1: Truong Nhat Quang (Student ID: DE180544)
 - Member 2: Nguyen Xuan Cuong (Student ID: DE180528)
- **Supervisor:** Ass.Prof Nguyen Luong Vuong

1.2. Project Overview

In the modern digital communication era, virtual keyboards act as the central interface connecting human thought with computer systems. For the Vietnamese language, traditional input methods like Telex and VNI require users to explicitly encode tone marks and diacritics. This project proposes an intelligent Vietnamese Input Method Editor (IME) that allows users to type a continuous sequence of Latin characters without accents or spaces (similar to the Pinyin input method for Chinese). Utilizing an end-to-end sequence-to-sequence architecture based on Deep Learning, the system automatically predicts the intended Vietnamese sentence with proper word boundaries and diacritics. The model is optimized for low-latency inference and designed to run entirely offline, preserving user privacy while providing a seamless typing experience.

2. Project Background

2.1. Overview of the Field

Vietnamese is an isolating language with a complex syllable structure consisting of an onset, rhyme, and one of six tone marks. The omission of diacritics and spaces introduces pervasive lexical ambiguity. The current Vietnamese typing methods require users to memorize explicit mapping rules (e.g., using 's', 'f', 'r', 'x', 'j' for tones in Telex) or switch to number rows (in VNI). While accurate, these methods impose a significant cognitive load and physical keystroke overhead.

2.2. Existing Systems

Existing systems range from traditional rule-based IMEs to advanced statistical models:

1. **Traditional IMEs (Telex, VNI):** Rely on deterministic keystroke composition. They lack contextual awareness and require high keystroke counts.
2. **Statistical and N-gram Models:** Earlier attempts at diacritic restoration used Hidden Markov Models (HMM) and Phrase-based Statistical Machine Translation (SMT). These suffer from the "curse of dimensionality" and fail on unsegmented continuous text.
3. **Recurrent Neural Networks (RNN, LSTM, GRU):** While Bi-LSTM and Bi-GRU models achieved good accuracy, their sequential processing nature introduces high inference latency, making them unsuitable for real-time keyboard applications.
4. **Modern Transformer Models (BARTpho, T5):** Pretrained Large Language Models have achieved state-of-the-art results in NLP tasks. However, their massive size makes offline mobile deployment challenging without significant optimization.

2.3. Limitations of Existing Solutions

- **High Cognitive Load and Physical Strain:** Traditional methods require excessive keystrokes (e.g., typing "tũi" requires 4 keys "tooi" in Telex) and explicit rule memorization.
- **Fragmented Processing:** Many existing AI solutions separate word segmentation and diacritic restoration, causing compounding errors.
- **Latency and Privacy Issues:** Cloud-based predictive models introduce network latency and pose severe privacy risks by transmitting user keystrokes to remote servers.

3. Problem Statement

Given a continuous sequence of unaccented Latin characters without whitespace (e.g., "toidihoc"), the system must jointly perform word segmentation and diacritic restoration to output the correct Vietnamese sentence (e.g., "tủi i hc" or "ti i hc"). This involves resolving massive homograph ambiguity and ambiguous word boundaries relying purely on global contextual dependency, while maintaining a strict inference latency (e.g., < 50 ms) suitable for real-time edge devices without internet connectivity.

4. Project Objectives

- **Dataset Construction:** Build a large-scale Vietnamese parallel corpus by converting standard text into unaccented, unsegmented sequences, including synthetic noise modeling for typo simulation.
- **Model Development:** Fine-tune a sequence-to-sequence Deep Learning model (based on the T5/ByT5 architecture) to jointly learn word boundaries and diacritics.
- **Inference Optimization:** Optimize the model for edge devices using techniques like Knowledge Distillation and Quantization to achieve low-latency offline execution.
- **Deployment:** Deploy the trained model as a fully offline backend service integrated with a custom interactive keyboard interface.

5. Significance of the Project

- **Enhanced Accessibility:** Removes the steep learning curve of Vietnamese typing for foreigners, children, and the elderly by acting as a natural language scaffold.
- **Improved Ergonomics:** Maximizes Keystroke Savings (estimated 20-40% reduction), significantly increasing typing speed and reducing the risk of Repetitive Strain Injury (RSI).
- **Absolute Privacy:** By executing entirely offline on the edge device, the system guarantees that sensitive user keystroke data is never exfiltrated to the cloud.

6. Project Scope & Limitations

- **In-Scope:** Developing the AI model, synthetic data pipeline, model optimization for edge inference, and a backend API demonstrating the continuous typing capabilities.

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- **Out-of-Scope:** OS-level low-level driver development for iOS/Android native integration, voice recognition, and handling of highly specialized ancient Vietnamese scripts.

II. Project Management Plan

1. Team Work

1.1. Team Structure and Roles

Role	Assignee	Core Responsibilities
Leader / AI Researcher	Tran Dinh Gia (SE181550)	• Lead the project and manage the schedule. • Formulate sequence modeling architectures and multi-task learning objectives. • Oversee model training and hyperparameter tuning.
Edge Systems & Model Optimization	Truong Nhat Quang (DE180544)	• Optimize decoding strategies (e.g., Beam Search). • Implement Knowledge Distillation and model quantization. • Benchmark inference latency and resource consumption on edge hardware.
Data Engineering & Backend Integration	Nguyen Xuan Cuong (DE180528)	• Construct and preprocess the large-scale Vietnamese parallel corpus. • Develop the data augmentation and synthetic noise generation pipelines. • Develop the offline inference backend and system integration APIs.

Table II..1: Team Structure, Roles, and Responsibilities

1.2. Communication Plan

- **Daily Standups (Asynchronous):** Conducted via messaging platforms to track daily commitments and identify technical blockers.
- **Weekly Supervisor Consultation:** Formal weekly meetings with Ass.Prof Nguyen Luong Vuong to review experimental results, validate model performance, and plan subsequent phases.
- **Version Control:** Centralized GitHub repository for code collaboration, utilizing Pull Requests and peer reviews.

2. Project Management Approach

2.1. Work Breakdown Structure

Phase	Milestone	Key Deliverables	Timeline
Phase 1	Literature Review & Data Pipeline	Vietnamese text corpora collection, synthetic noise generation (diacritic stripping, space erasure), tokenization setup.	Weeks 1–4
Phase 2	Model Development & Training	Fine-tuning pretrained sequence-to-sequence model (T5-based), hyperparameter optimization, error analysis.	Weeks 5–9
Phase 3	Model Optimization & Evaluation	Decoding optimization, knowledge distillation, quantization, benchmarking BLEU, WER, and latency metrics.	Weeks 10–14
Phase 4	Backend & System Integration	Development of offline inference backend, API design for keyboard interaction, functional testing.	Weeks 15–17
Phase 5	Validation & Deployment	Comprehensive user testing, Keystroke Savings evaluation, final documentation, and project defense preparation.	Weeks 18–20

Table II.2: Work Breakdown Structure and Project Timeline

2.2. Risk Management

Identified Risk	Sev.	Prob.	Impact	Mitigation Strategy
Domain Generalization Gap	High	Med	High accuracy on training data but poor performance on noisy real-world typing.	Include synthetic typo noise (fat-finger errors) and non-standard words (teencode) during training.
High Inference Latency	High	Med	Model responds too slowly (> 50ms), disrupting the typing experience.	Aggressively apply Knowledge Distillation and INT8 quantization; utilize lightweight Transformer variants.
Compute Resource Limits	Med	High	Slow training iterations due to large model sizes and extensive dataset.	Leverage distributed cloud GPUs and mixed-precision training (FP16).

Table II.3: Risk Register and Mitigation Strategies

2.3. Quality Management

- **Linguistic Accuracy Metrics:** Evaluate using Word Error Rate (WER), Character Error Rate (CER), Exact Match ratio, and BLEU Score.
- **Performance Metrics:** Ensure inference latency remains under 50ms per keystroke event for seamless interaction.

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- **Ergonomic Metrics:** Measure Keystroke Savings (KS) to quantify the reduction in physical typing effort compared to standard Telex.

2.4. Budget and Funding (Optional)

- **Infrastructure:** Utilizing free-tier or academic allocations on platforms like Kaggle, Google Colab, and local workstations.
- **Total Budget:** \$0.00 USD.

2.5. Change Management Process

- **Architecture Pivot:** If a chosen model architecture fails to meet latency requirements by Phase 3, the team will pivot to a smaller predefined baseline (e.g., CNN-LSTM hybrid).
- **Scope Adjustment:** Secondary features like personalized user dictionaries will be deprioritized if the core continuous typing engine requires extended development time.

2.6. Closure and Evaluation

- **Deliverables Handover:** Source code, trained model weights, API documentation, and evaluation reports.
- **Final Thesis:** A comprehensive academic report detailing the methodology, empirical results, and system architecture.

III. Methodology

[Provide the final methodology following the template as part II in the Report #4]

[This section will detail how you will approach the project, with a focus on applying and implementing AI techniques. The team is allowed to customize the content of this section, however, the team needs to provide a detailed view of the methods and techniques used. To ensure that the reader understands the scientific and systematic approach that you apply to your capstone project in the field of AI. Below are two sample templates for this section that you can choose from or customize.]

1. Overview of Framework/Approach

[Describe the whole system in general.]

2. Data Collection and Preprocessing

[Describe how you will collect data, including the sources of data, methods of collection, and any selection criteria. Information about the sample size, diversity, and representativeness of the data should be included.]

[Detail how the data will be processed before analysis, including cleaning data, handling missing data, transforming data, and normalization.]

3. Feature Selection and Engineering

[Describe the process of selecting and creating features that will be used for the machine learning model, as well as the reasons for these choices.]

4. Proposed Model/System/Method (Contributions)

[Describe the contributions of the capstone project in detail.]

5. Experimental Settings

[Describe the machine learning models considered and the reasons for the final model choice, including technical specifications and configurations of the model.]

[Detail how the model will be trained and validated, including information on data splitting methods, cross-validation techniques, and the metrics used for evaluation.]

6. Evaluation Metrics

[Identify the metrics that you will use to assess the performance of the model, such as accuracy, F1 score, ROC AUC, etc.]

IV. Results and Discussion

[Provide the final results and discussion following the template as part II in the Report #6]

1. Results and Analysis

[Present the results obtained from executing the project, including numerical data, graphs, diagrams, and supportive imagery. This may include the outcomes of machine learning models, such as accuracy, confusion matrices, ROC curves, and other evaluation metrics.]

[Provide a detailed analysis of the results obtained, explaining their significance and impact on the project. This might include comparing the performance of different models, error analysis, and interpreting the features of the model.]

2. Discussion

[Discuss the results, including how they address the research questions and hypotheses, their alignment with results from related studies, and the broader context of the project. Mention any limitations or challenges encountered during the project.]

3. Personal Reflection (Optional)

[A section reflecting personally on the project process, lessons learned, and how improvements could be made in the future.]

[Note: The "Results and Discussion" section is an essential part of the capstone project document, where you evaluate the success of your project based on data and analysis. This is also an opportunity for you to reflect and discuss the larger significance of the work you have done, as well as to identify any opportunities for further research or development.]

V. System Design and Implementation

[Provide the final system design and implementation following the template as part II in the Report #5]

1. AI Model Integration

[Describe how the AI model is integrated into the larger system, including the interface and data exchange between the AI model and other system components.]

2. Data Flow and Processing

[Illustrate and describe the data flow through the system, from data collection, processing, to data analysis and result presentation.]

3. Deployment Strategy

[Detail how the system will be deployed, including the deployment environment and deployment procedures.]

4. Scalability and Maintenance

[Discuss how the system is designed to handle growth and future maintenance, including system updates, scalability, and version management.]

[Note: By providing a detailed view of the AI solution's design and implementation in this section, you will help readers understand the technical and systematic approach you have applied to your capstone project.]

VI. Conclusion

[Provide the final conclusion following the template as part II in the Report #7]

1. Summary of Findings

[Summarize the main findings of the project, including how the research questions have been answered and how the project goals were achieved.]

2. Contributions and Reflections on the Project

[Highlight the specific contributions of the project to the field of study, community, or industry. This could include technical advancements, new knowledge, or practical applications.] [Reflect on the project process, including what worked well and what could be improved. This is also an opportunity to acknowledge the level of challenge and learning from the project.] [Provide some final thoughts or conclusion, expressing hopes or vision for the future of the research or the problem addressed by the project.]

3. Limitations and Future Work

[Acknowledge any limitations of the project and how they may affect the results and conclusions. This includes factors such as data, methods, or assumptions.] [Suggest directions for future research or development based on the results and limitations of the project. This helps guide others who may build upon your work.]

[Note: The "Conclusion and Future Work" section is not only a place to summarize the completed work but also an opportunity to highlight the importance of the project and propose next steps. This helps readers understand the impact and potential of the work you have done.]

VII. Ethical Declaration

1. Ethical Considerations

[Discuss ethical issues that may arise when applying machine learning, including data privacy, accuracy, and how to handle data fairly.]

2. Declaration of AI-based Assistance

[Declare AI-based tools that have been used during doing the capstone project in detail.]

VIII. Appendices

This section can be removed out of the report if the capstone project does not contain any appendices.

1. First Appendix Sample

[Provide the final appendices following the template as part II in the Report #9]

[If there is any supplementary information that does not fit into the main sections of the document but is still important and useful for a full understanding of the project, you can include it in the appendices. This might include raw data, source code, full versions of surveys, more detailed diagrams, or other supporting documents for your main content.]

[Note: After including all these sections, ensure that all parts of the document are logically and coherently organized, and that you have thoroughly checked for language, spelling, and grammar errors.]

2. Second Appendix Sample

[Note: After including all these sections, ensure that all parts of the document are logically and coherently organized, and that you have thoroughly checked for language, spelling, and grammar errors.]

This part presents the instructions of the format of the references.

To manage the references, some tools, such as mendeley or jabref, have been recommended because the reference will only appear in referecne list once it is cited in the report. Try it!

You have to cite at least 1 paper, such as LSTM [1], Fast R-CNN [2], Transformer [3] or EfficientDet [4], if IEEE style \bibliographystyleIEEEtran and its bibliography style \bibliographyreferences have been used for managing and citing references.

[Provide the final references following the template as part II in the Report #8]

[Formatting: Ensure that all references are listed in a uniform and correct format, following the guidelines of the citation system you have chosen (e.g., APA, MLA, Chicago, etc.). This helps ensure accuracy and ease of source verification.]

[Order: Typically, reference items are arranged in alphabetical order for easy lookup. Make sure that all sources are listed comprehensively and without omission.]

[Completeness: For each reference, provide enough information so that readers can find and review the source material. This includes the author's name, year of publication, title of the work, publisher name or journal name, and other relevant information such as page numbers or URL.]

[Accuracy: Carefully check to ensure that all information about the references is correct and up-to-date.] [Relevance: Only include sources that were actually cited or used in your project. This helps maintain the integrity and accuracy of the project.]

[Note: The "References" section is an indispensable part of any academic project. It not only proves that you have conducted careful and grounded research but also allows future generations of students to check and expand their projects from the sources you have cited.]

[You should remove this part out of the final manuscript.]

IX. References

- [1] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
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